

Module/Pathway/Taxon Review: L-aspartate de novo and selected salvage routes to NAD⁺ in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Target bucket:** KEGG ppu00760 — Nicotinate and nicotinamide metabolism **Module area:** cofactors_vitamins_redox **Scope:** L-aspartate de novo NAD⁺ synthesis plus selected nicotinate / nicotinamide / NMN salvage routes to NAD⁺ (kynurenine/tryptophan de novo route explicitly out of scope; owned by MODULE:kynurenine_nad_de_novo)

1. Executive summary

In *Pseudomonas putida* KT2440 the aspartate-de-novo-and-salvage NAD⁺ module is **satisfiable end-to-end**. The organism encodes a complete, single-copy L-aspartate de novo route to nicotinate mononucleotide (NaMN) — **NadB** (L-aspartate oxidase, PP_1426), **NadA** (quinolinate synthase, PP_1231), and **NadC** (quinolinate phosphoribosyltransferase, PP_0787) — which converges with two functional salvage entry points, the nicotinate Preiss–Handler enzyme **PncB** (PP_4868) and the NMN-deamidase **PncC** (PP_1628), onto a shared NaMN → deamido-NAD → NAD⁺ completion carried by **NadD** (NaMN adenylyltransferase, PP_4810, bacterial NadD family) and **NadE** (NH₃-dependent NAD synthetase, PP_4869). Every core biosynthetic step of the module is covered by a clearly annotated candidate gene whose EC number matches the expected activity, and each core step is single-copy — there is little paralog ambiguity in the biosynthetic backbone.

The one genuine **gap** is the nicotinamide-deamidation salvage branch. KT2440 has **no canonical nicotinamidase (PncA, EC 3.5.1.19)**: a proteome-wide EC and name search of UP000000556 returns no PncA, and the only member of the parent isochorismatase superfamily is a single uncharacterized protein (Q88JT5). Consequently, the deamidating route from nicotinamide (Nm → nicotinate → NaMN via PncA+PncB) cannot currently be marked

covered. Two module *variants* are **not expected** in this organism at all: the NAMPT-to-direct-NMN route (no EC 2.4.2.12, no NadM/NMNAT EC 2.7.7.1) and the glutamine-dependent NAD-synthetase architecture (the resident NadE is ammonia-dependent, EC 6.3.1.5).

Finally, the KEGG bucket ppu00760 is a **poor proxy for the module boundary**: of the 30 candidate genes, only ~8 lie inside the aspartate-de-novo/salvage-to-NAD⁺ scope. The remaining ~22 are out-of-scope over-inclusions — the entire aerobic nicotinate *degradation* cluster (nicAB/nicCDEFX/maiA), pyridine-nucleotide transhydrogenases (pntAA/pntB/pntAB/sthA), NAD kinase (nadK), NAD-consuming enzymes (cobB sirtuin, nudC NAD-cap hydrolase), promiscuous 5'-nucleotidases (ushA/surE/PP_2531), and succinate/glutarate-semialdehyde dehydrogenases that actually belong to ppu00350. This is a **bucket-level over-annotation problem**, not a module defect.

2. Target-organism pathway definition

What is included

The module covers the routes that **produce NAD⁺** from (a) the L-aspartate de novo pathway and (b) selected pyridine salvage precursors (nicotinate, nicotinamide, NMN), through the universal downstream completion NaMN → deamido-NAD (NaAD) → NAD⁺. Concretely, for KT2440 the in-scope chemistry is:

- **De novo**: L-aspartate → iminoaspartate (NadB) → quinolinate (NadA) → NaMN (NadC, with decarboxylation and PRPP condensation).
- **Preiss-Handler salvage**: nicotinate + PRPP → NaMN (PncB).
- **NMN-deamidation salvage**: NMN → NaMN (PncC).
- **Shared completion**: NaMN + ATP → deamido-NAD (NadD); deamido-NAD + NH₃ → NAD⁺ (NadE).

What must be kept separate

- **Tryptophan/kynurenine de novo route** — owned by MODULE:kynurenine_nad_de_novo; not duplicated here.
- **Nicotinate catabolism / degradation** — the aerobic NA degradation cluster (nic genes) channels nicotinate to central metabolism and is a **separate catabolic module**, even though several nic genes carry the ppu00760 KEGG tag ([PMID: 18678916](#), [PMID: 21450002](#)).

- **Pyridine-nucleotide transhydrogenases** (pntAB, sthA), **NAD kinase / NADP formation** (nadK), **NAD-consuming reactions** (cobB deacylase, nudC NAD-cap hydrolase), and **unrelated aldehyde metabolism** (semialdehyde dehydrogenases). These are explicitly outside the module boundary in the working scope.

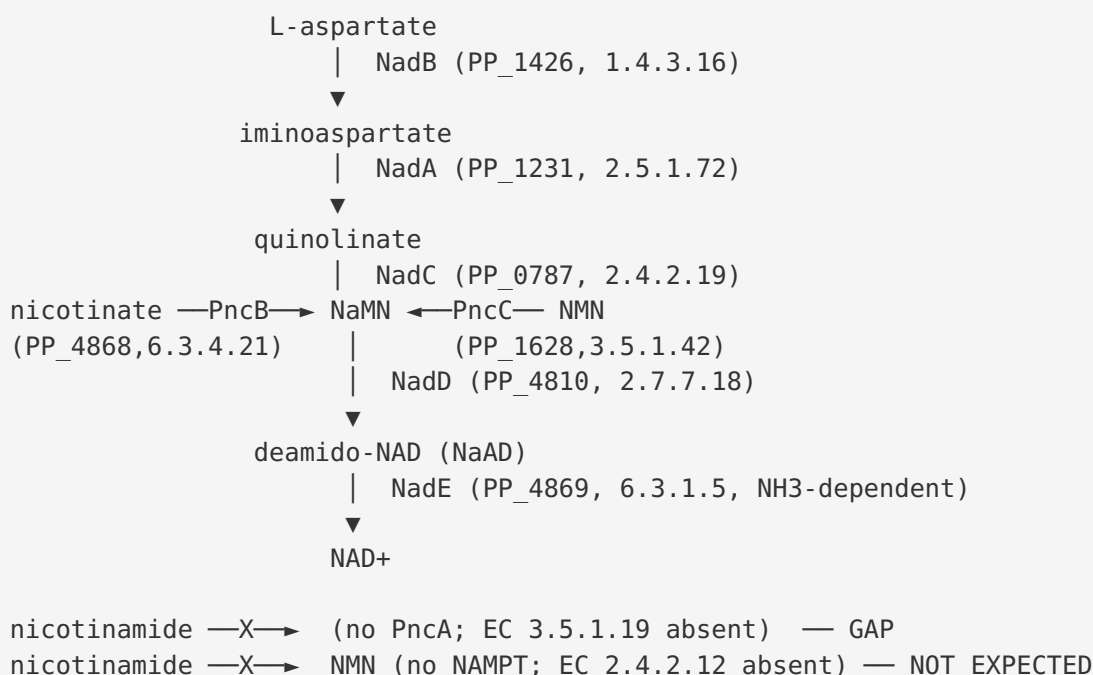
Alternate names / database definitions

The KEGG map ppu00760 "Nicotinate and nicotinamide metabolism" is a broad overview map that mixes biosynthesis, salvage, and degradation. The MetaCyc/BioCyc equivalents split these into **NAD biosynthesis I (from aspartate)**, **NAD salvage pathway**, and **nicotinate degradation**. The module reviewed here corresponds to the biosynthesis + salvage-to-NAD+ subset only. Curators should not treat "gene tagged ppu00760" as equivalent to "gene in this module."

3. Expected step model

The table below is the module's expected step model instantiated for KT2440.

Step	Expected activity (EC)	Enzyme family	KT2440 gene	Call
De novo 1: L-aspartate oxidation	L-aspartate oxidase (1.4.3.16)	NadB	PP_1426 / Q88MZ2	covered
De novo 2: quinolinate formation	quinolinate synthase (2.5.1.72)	NadA	PP_1231 / Q88NH8	covered
De novo 3: quinolinate → NaMN	NaMN pyrophosphorylase, carboxylating (2.4.2.19)	NadC	PP_0787 / Q88PR1	covered
Salvage: nicotinate → NaMN	nicotinate PRTase (6.3.4.21 / 2.4.2.11)	PncB	PP_4868 / Q88DF7	covered
Salvage: Nm → nicotinate	nicotinamidase (3.5.1.19)	PncA	—	gap / candidate_uncertain
Salvage: NMN → NaMN	NMN amidohydrolase (3.5.1.42)	PncC	PP_1628 / Q88ME5	covered
Completion 1: NaMN → deamido-NAD	NaMN adenylyltransferase (2.7.7.18)	NadD (bacterial)	PP_4810 / Q88DL5	covered
Completion 2: deamido-NAD → NAD ⁺	NAD synthetase, NH ₃ -dependent (6.3.1.5)	NadE	PP_4869 / Q88DF6	covered
Variant: NAMPT direct route	Nm phosphoribosyltransferase (2.4.2.12)	NAMPT/ NadV	—	not_expected
Variant: direct NMN adenylylation	NMN adenylyltransferase (2.7.7.1)	NadM/ NMNAT	—	not_expected
Variant: Gln-dependent completion	NAD synthetase, Gln-hydrolyzing (6.3.5.1)	NADSYN1-type	—	not_expected (resident NadE is NH ₃ -dependent)



4. Candidate genes and evidence

4.1 High-confidence, in-scope core genes (covered)

NadB — PP_1426 / Q88MZ2 — L-aspartate oxidase (EC 1.4.3.16). First committed step of de novo synthesis. UniProt EC annotation matches the expected activity exactly; single copy. Evidence type: sequence/EC annotation in the target proteome, strong homology to characterized enterobacterial NadB. Curation call: **covered**, no caveat.

NadA — PP_1231 / Q88NH8 — quinolinate synthase (EC 2.5.1.72). Condenses iminoaspartate + DHAP to quinolinate; carries a [4Fe-4S] cluster in characterized homologs. EC match exact, single copy. Curation call: **covered**.

NadC — PP_0787 / Q88PR1 — quinolinate phosphoribosyltransferase / NaMN pyrophosphorylase, carboxylating (EC 2.4.2.19). Converts quinolinate + PRPP to NaMN. This is the convergence node where de novo synthesis meets the shared completion. Note the cross-species precedent that NadC can also function in **quinolinate salvage** even where the rest of de novo synthesis is absent ([PMID: 23204464](#)); in KT2440 the full de novo route is present, so NadC is the terminal de novo step. Curation call: **covered**.

PncB — PP_4868 / Q88DF7 — nicotinate phosphoribosyltransferase (EC 6.3.4.21). Preiss-Handler entry: nicotinate + PRPP → NaMN (ATP-dependent NAPRTase). Physically adjacent to nadE (PP_4869), a common bacterial salvage/completion gene arrangement. Curation call: **covered** for the nicotinate salvage branch.

PncC — PP_1628 / Q88ME5 — NMN amidohydrolase / nicotinamide-nucleotide amidase (EC 3.5.1.42). Deamidates NMN to NaMN, feeding NMN back into the deamido-NAD completion. Member of the PncC/CinA C-terminal amidase family. Curation call: **covered** for the NMN-deamidation branch. Caveat: PncC provides only an *indirect* nicotinamide-salvage capacity (Nm would still need to first become NMN), so it does not substitute for a missing PncA in the classical Nm → nicotinate deamidation step.

NadD — PP_4810 / Q88DL5 — NaMN adenylyltransferase (EC 2.7.7.18), bacterial NadD family. Adenylylates NaMN to deamido-NAD; the module's "Bacterial NadD-family" variant, not the eukaryotic bifunctional NMNAT variant. This and NadE are the two indispensable downstream enzymes of NAD synthesis and validated antibacterial targets in other bacteria ([PMID: 23204464](#), [PMID: 20926389](#)). Curation call: **covered**.

NadE — PP_4869 / Q88DF6 — NH₃-dependent NAD synthetase (EC 6.3.1.5). Amidates deamido-NAD to NAD⁺. The UniProt annotation explicitly specifies the **ammonia-dependent** subfamily, which resolves the module's nitrogen-donor variant question: KT2440 uses the ammonia-dependent architecture, **not** the glutamine-hydrolyzing NADSYN1-type ([PMID: 29581233](#)). Curation call: **covered**; glutamine-dependent variant **not_expected**.

4.2 In-bucket but out-of-scope genes (should NOT count toward module satisfiability)

Gene(s)	Annotation / EC	Why out of scope
pntB, pntAA, pntAB (PP_0155/0156/5747), sthA (PP_2151)	transhydrogenases (7.1.1.1; 1.6.1.1)	interconvert NAD(H)/ NADP(H); not NAD+ <i>biosynthesis</i>
nadK (PP_2012)	NAD kinase (2.7.1.23)	makes NADP from NAD+; downstream of module boundary
cobB (PP_5402)	NAD-dependent deacylase / sirtuin (2.3.1.286)	NAD- consuming , not biosynthetic
nudC (PP_4029)	NAD-cap RNA hydrolase / NADH pyrophosphatase (3.6.1.22)	NAD turnover/decapping
mazG (PP_1657)	NTP pyrophosphohydrolase (3.6.1.8)	primary bucket is ppu00770; house-cleaning
ushA, surE, PP_2531	5'-nucleotidases (3.1.3.5)	promiscuous phosphatases; not committed NAD steps
nicAB, nicC, nicD, nicF, nicX, maiA (PP_3941–3948)	nicotinate degradation cluster	catabolic; separate module (PMID: 18678916)
davD, sad-I, sad-II, gabD-II	semialdehyde dehydrogenases (1.2.1.-/24/79)	belong to ppu00350; unrelated aldehyde metabolism
PP_3103	uncharacterized protein	no assigned activity; unclear relevance

5. Gaps, ambiguities, and likely over-annotations

5.1 The nicotinamide-deamidation gap (PncA)

A proteome-wide SPARQL search of UP000000556 / taxon 160488 returned **no protein carrying EC 3.5.1.19** (nicotinamidase/PncA) and no protein named "nicotinamidase" or "PncA." The classical deamidating nicotinamide-salvage branch ($\text{Nm} \rightarrow \text{nicotinate} \rightarrow \text{NaMN}$, via PncA then PncB) is therefore **not encoded by a canonical gene**. In comparative studies, PncA is the enzyme that specifically enables **deamidating** utilization of nicotinamide, distinct from the nondeamidating NAMPT/NadV route ([PMID: 20926389](#)), and its essentiality for Nm utilization has been shown directly in other bacteria ([PMID: 23204464](#)). Its absence in KT2440 removes the deamidating Nm route while leaving the nicotinate (PncB) branch intact.

The **only cryptic candidate** is Q88JT5, a single "isochorismatase-like domain-containing protein" — the isochorismatase superfamily is the fold that houses PncA — but it is uncharacterized and its EC is unassigned. This is a weak, fold-only lead. Curation call for the Nm-deamidation step: **gap / candidate_uncertain**, with Q88JT5 as the sole promotion candidate.

5.2 Precursor uptake ambiguity

No protein is annotated as a **nicotinate/niacin transporter (NiaP)** by name in the proteome. The only NAD-precursor transporter present is **PnuC (Q88IY9)**, a nicotinamide-riboside transporter, together with an **NadR/Ttd14-like AAA-domain protein (Q88IZ0)** — but the latter is *not* a classical bifunctional NadR NMNAT. Thus even the salvage precursor-uptake picture is incomplete/ambiguous, which is relevant when assessing whether salvage branches are physiologically operative versus merely genetically latent.

5.3 Variants not expected in KT2440

- **NAMPT/NadV direct route** ($\text{Nm} \rightarrow \text{NMN}$, EC 2.4.2.12): **absent** — no EC 2.4.2.12 in the proteome. → **not_expected_in_target_taxon**.
- **NadM/NMNAT direct NMN adenylylation** ($\text{NMN} \rightarrow \text{NAD}^+$, EC 2.7.7.1): **absent** — this is the alternative "amidate-then-adenylylate" route characterized in *Francisella tularensis* ([PMID: 19204287](#)), and KT2440 lacks it. → **not_expected_in_target_taxon**.
- **Glutamine-dependent NAD synthetase**: resident NadE is NH_3 -dependent → glutamine-dependent variant **not_expected**.

5.4 Bucket-level over-annotation

The dominant over-annotation issue is at the **KEGG bucket level**: ppu00760 sweeps in degradation, transhydrogenation, NADP formation, NAD consumption, and unrelated aldehyde metabolism. ~22 of 30 candidates are out of scope. This is a systematic mismatch between a broad KEGG overview map and a mechanistically-scoped biosynthesis+salvage module — a **module_needs_revision flag against the bucket→module mapping**, not against the module content itself.

6. Module and GO-curation recommendations

Per-step module status calls:

Module step	Status
De novo: L-aspartate oxidase (NadB)	covered
De novo: quinolinate synthase (NadA)	covered
De novo: quinolinate PRTase (NadC)	covered
Preiss–Handler: PncB	covered
NMN deamidation: PncC	covered
NaMN adenylation: NadD (bacterial family)	covered
Deamido-NAD amidation: NadE (NH ₃ -dependent)	covered
Nm deamidation: PncA	gap / candidate_uncertain
NAMPT direct NMN production	not_expected_in_target_taxon
Direct NMN adenylation (NadM/NMNAT)	not_expected_in_target_taxon
Gln-dependent NAD synthetase variant	not_expected_in_target_taxon

Overall module verdict: SATISFIABLE. A complete route from L-aspartate to NAD⁺ exists (de novo → NadD → NadE), and two independent salvage entries (PncB, PncC) also converge on the same completion. Module satisfiability does not depend on the missing PncA branch.

Boundary corrections: The generic module boundaries are **correct for this organism**; it is the KEGG **ppu00760 → module bucket mapping** that is wrong. Recommend that curation: 1. Restrict the module's KT2440 gene set to the ~8 in-scope genes (nadB, nadA, nadC, nadD, nadE, pncB, pncC; plus transporter context). 2. Reassign nic-cluster genes to a nicotinate-degradation module, semialdehyde dehydrogenases to ppu00350, and transhydrogenase/NAD-kinase/NAD-consuming genes to their own redox/turnover modules.

GO/annotation actions: No new GO term appears necessary — all covered activities map to existing EC/GO terms. Flag Q88JT5 for possible future annotation if experimentally shown to be a nicotinamidase. No new module document is needed; the existing module is adequate once the bucket over-inclusion is pruned.

7. Genes to promote to full fetch-gene review

1. **Q88JT5 (isochorismatase-like domain protein)** — highest priority. Sole fold-level candidate for the missing PncA/nicotinamidase step. Full review should assess active-site residues (the Cys-Asp-Lys catalytic triad of nicotinamidases), operon context, and any phenotypic/expression data. Resolving this decides whether the Nm-deamidation step is a true gap or a cryptic assignment.
 2. **Q88IZ0 (NadR/Ttd14-like AAA-domain protein)** — clarify whether it retains any NMN adenylyltransferase / salvage-regulatory function, given it is *not* a classical bifunctional NadR.
 3. **PncC — PP_1628** — confirm that the annotation reflects genuine NMN amidohydrolase activity versus the regulatory CinA-associated role, since PncC/CinA-family proteins are bifunctional in some taxa.
 4. **NadE — PP_4869** — verify the ammonia-dependent subfamily assignment (vs. any latent glutaminase capacity), as this fixes the nitrogen-donor variant call.
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8. Limitations and knowledge gaps

- **Absence calls rest on negative searches.** No-hit results for PncA (EC 3.5.1.19), NAMPT (EC 2.4.2.12), and NadM (EC 2.7.7.1) are strong but not absolute: a diverged or mis-annotated

ORF could evade EC-based matching. This is exactly why Q88JT5 is flagged for full review rather than the gap being declared final.

- **No direct KT2440 enzymology was reviewed for the biosynthetic backbone.** The covered calls are based on target-strain proteome annotation (UniProt UP000000556, EC-matched, single-copy) and strong homology, not on strain-specific in vitro assays. This is standard and adequate for curation but is homology/annotation evidence, not direct experiment.
- **Salvage physiology is uncertain.** The incomplete transporter annotation (only PnuC named; no NiaP nicotinate permease) means we cannot confirm from genetics alone whether the PncB/PncC salvage branches are physiologically active under given conditions or merely latent.
- **Species transfer.** Mechanistic and target-relevance context is transferred from *Acinetobacter*, group A streptococci, *Francisella*, and *Bordetella*. Transfer is **strong** for the conserved downstream completion enzymes (NadD/NadE) and **moderate** for salvage-branch physiology.

9. Proposed follow-up experiments / curator actions

1. **Characterize Q88JT5** — heterologous expression + nicotinamidase assay (Nm → nicotinate + NH₃) to accept or reject it as the missing PncA. Highest-value single experiment.
2. **Growth-phenotype panel** — test KT2440 growth on nicotinamide vs. nicotinate vs. quinolinate as sole pyridine source (and in de novo-blocked backgrounds) to confirm which salvage entries are functional in vivo.
3. **Prune the ppu00760 bucket→module mapping** — reassign the ~22 out-of-scope genes to degradation/redox/turnover modules; retain only nadB, nadA, nadC, nadD, nadE, pncB, pncC (+ transporter context) for this module.
4. **Confirm NadE nitrogen donor** — verify ammonia- vs glutamine-dependence to finalize the nitrogen-donor variant call.
5. **Search for a NiaP-type nicotinate transporter** by profile/HMM to close the precursor-uptake gap.

10. Key references

- **Nicotinate degradation cluster in KT2440** — *Deciphering the genetic determinants for aerobic nicotinic acid degradation: the nic cluster from Pseudomonas putida KT2440*. PMID: [18678916](#). Establishes the nic genes as **catabolic**, supporting their exclusion from the biosynthesis module.
 - **Regulation of the nic pathway** — *A finely tuned regulatory circuit of the nicotinic acid degradation pathway in Pseudomonas putida*. PMID: [21450002](#). Confirms nicAB / nicCDEFTP / nicXR operon structure and degradation role; notes the link to preserving vitamin B3 for cofactor synthesis.
 - **PncA role in deamidating Nm salvage** — *Genomics-driven reconstruction of acinetobacter NAD metabolism*. PMID: [20926389](#). Distinguishes the deamidating (PncA) and nondeamidating (NadV/NAMPT) nicotinamide routes — basis for calling the KT2440 PncA branch a gap.
 - **PncA essentiality; NadC quinolinate salvage; NadD/NadE as targets** — *Quinolinate salvage and insights for targeting NAD biosynthesis in group A streptococci*. PMID: [23204464](#). Supports the interpretation of NadC and identifies NadD/NadE as the indispensable downstream completion enzymes.
 - **Alternative NaMN→NMN→NAD route** — *Nicotinamide mononucleotide synthetase is the key enzyme for an alternative route of NAD biosynthesis in Francisella tularensis*. PMID: [19204287](#). Defines the NadM/NMNAT variant that KT2440 lacks (not_expected).
 - **NAD synthetase nitrogen-donor architectures** — *Kinetics and structural features of dimeric glutamine-dependent bacterial NAD synthetase*. PMID: [29581233](#). Context for the ammonia- vs glutamine-dependent NadE variant distinction.
 - **Cross-taxon nic locus conservation** — *The Bordetella bronchiseptica nic locus encodes a nicotinic acid degradation pathway... highly similar to that characterized in Pseudomonas putida KT2440*. PMID: [29485696](#). Corroborates that the KT2440 nic locus is a degradation system that nonetheless influences NAD precursor pools.
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Evidence provenance note

Core gene calls (NadB, NadA, NadC, NadD, NadE, PncB, PncC) rest on **direct proteome evidence for the target strain** (UniProt UP000000556 / taxon 160488, EC-number matched, single-copy). Absence calls (PncA, NAMPT, NadM) rest on **negative proteome-wide EC/name**

searches — strong but not absolute. Mechanistic roles and target relevance are transferred from related bacteria; this transfer is **strong for conserved downstream completion enzymes** but only **moderate for salvage-branch physiology**, since precursor-transport annotation in KT2440 is incomplete.

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